Numerical methods

Lecture 5: Matrix norms, sensitivity of linear systems

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Definition: matrix norm

Fix $n \in \mathbb{N}$. The map $\|.\| : \mathbb{R}^{n\times n} \to \mathbb{R}$ is called a *matrix norm* if:

1. $\|A\| \geq 0 \quad (\forall A \in \mathbb{R}^{n\times n})$,
2. $\|A\| = 0 \iff A = 0$,
3. $\|\lambda \cdot A\| = |\lambda| \cdot \|A\| \quad (\forall \lambda \in \mathbb{R},\ \forall A \in \mathbb{R}^{n\times n})$,
4. $\|A + B\| \leq \|A\| + \|B\| \quad (\forall A, B \in \mathbb{R}^{n\times n})$,
5. $\|A \cdot B\| \leq \|A\| \cdot \|B\| \quad (\forall A, B \in \mathbb{R}^{n\times n})$.

Same as in the case of vector norms, plus: 'submultiplicativity'.
These are the *axioms* of matrix norms.

Definition: Frobenius norm

The below function is called the *Frobenius norm:*

$$\|.\|_F : \mathbb{R}^{n\times n} \to \mathbb{R}, \qquad \|A\|_F = \left(\sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}|^2\right)^{1/2}.$$

Proposition: Frobenius norm

The $\|.\|_F$ function is indeed a matrix norm.

Proof. 1–4 follows from the properties of vector norm $\|.\|_2$.
The 5th can be proved using CBS. $\square$

Example: Frobenius norm

Find the Frobenius norm of the below matrices.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix}.$$

$$\|A\|_F = \sqrt{1^2 + (-4)^2 + 2^2 + 2^2} = 5$$

$$\|B\|_F = \sqrt{3^2 + 2^2 + 1^2 + 5^2} = 6$$

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Definition: induced norms, natural matrix norms

Let $\|.\|_v : \mathbb{R}^n \to \mathbb{R}$ be a vector norm. The function

$$\|.\| : \mathbb{R}^{n\times n} \to \mathbb{R}, \qquad \|A\| := \sup_{x\neq 0} \frac{\|Ax\|_v}{\|x\|_v}$$

is called the *matrix norm induced by* $\|.\|_v$. A matrix norm is called *natural*, if it is induced by a vector norm.

Theorem: induced norms

The 'induced matrix norms' are indeed matrix norms.

Natural matrix norms, induced norms

Proof. It is to show that the given form satisfies the axioms of matrix norms.

1. The value of $\|A\|$ is non-negative, since it is a supremum of a set of fractions of vector norms (non-negativ numbers).
2. If $A = 0$, i.e. null matrix, then $\|Ax\|_v = 0$ for all x vectors, thus the supremum is also 0. And conversely: if the supremum is 0, then Ax must be zero vector for all x vectors; thus A must be a null matrix.
3. $$\|\lambda A\| = \sup_{x\neq 0} \frac{\|\lambda Ax\|_v}{\|x\|_v} = \sup_{x\neq 0} \frac{|\lambda| \cdot \|Ax\|_v}{\|x\|_v} = |\lambda| \cdot \sup_{x\neq 0} \frac{\|Ax\|_v}{\|x\|_v} = |\lambda| \cdot \|A\| .$$

Proof continued.

4.

$$\|A+B\| = \sup_{x\neq 0} \frac{\|(A+B)x\|_v}{\|x\|_v} \leq \sup_{x\neq 0} \frac{\|Ax\|_v + \|Bx\|_v}{\|x\|_v} \leq$$

$$\leq \sup_{x\neq 0} \frac{\|Ax\|_v}{\|x\|_v} + \sup_{x\neq 0} \frac{\|Bx\|_v}{\|x\|_v} = \|A\| + \|B\|$$

5. $B = 0 \;\; \Rightarrow \;\; \|B\| = 0$, and $A \cdot B = A \cdot 0 = 0 \;\; \Rightarrow \;\; \|AB\| = 0$.
Both sides are 0 thus the statement is true.

Proof continued. If $B \neq 0$, then

$$\|A \cdot B\| = \sup_{x \neq 0} \frac{\|ABx\|_v}{\|x\|_v} = \sup_{x \neq 0, Bx \neq 0} \frac{\|ABx\|_v}{\|Bx\|_v} \cdot \frac{\|Bx\|_v}{\|x\|_v} \leq$$

$$\leq \sup_{Bx \neq 0} \frac{\|ABx\|_v}{\|Bx\|_v} \cdot \sup_{x \neq 0} \frac{\|Bx\|_v}{\|x\|_v} \leq \sup_{y \neq 0} \frac{\|Ay\|_v}{\|y\|_v} \cdot \sup_{x \neq 0} \frac{\|Bx\|_v}{\|x\|_v} = \|A\| \cdot \|B\| \, .$$

Consider, that the condition $Bx \neq 0$ does not change the value of the supremum; the notation $y := Bx$ was introduced. $\square$

Remarks:

- An alternate form:

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|_v}{\|x\|_v} = \sup_{\|y\|_v = 1} \|Ay\|_v \, .$$

- One can write max instead of sup.
- Yet another form:

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|_v}{\|x\|_v} \implies \frac{\|Ax\|_v}{\|x\|_v} \leq \|A\| \implies \|Ax\|_v \leq \|A\| \cdot \|x\|_v \, .$$

  Furthermore $\|A\|$ is the least upper bound.

Natural matrix norms, induced norms

Definition: consistent norms

If for a matrix norm and for a vector norm

$$\|Ax\|_v \leq \|A\|_m \cdot \|x\|_v \qquad (\forall x \in \mathbb{R}^n,\ A \in \mathbb{R}^{n\times n})$$

holds, then they are called *consistent*.

Proposition: the consistency of induced norms

Natural norms are consistent with the vector norm they are induced by.

Proof. We just saw that. (Consider $x = 0$ too.) $\square$

Natural matrix norms, induced norms

What matrix norms do the common vector norms induce?

Theorem: Common matrix norms $(1, 2, \infty)$

The matrix norms induced by the vector norms $\|.\|_p$ $(p = 1, 2, \infty)$:

- $\|A\|_1 = \max_{j=1}^{n} \sum_{i=1}^{n} |a_{ij}|$ (column norm),
- $\|A\|_\infty = \max_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|$ (row norm),
- $\|A\|_2 = \left(\max_{i=1}^{n} \lambda_i(A^\top A) \right)^{1/2}$ (spectral norm).

Notation: $\lambda_i(M)$: the ith eigenvalue of the matrix M $(Mv = \lambda v,\ v \neq 0)$.

Outline of the proof(s):

- For the stated value $f(A)$: $\|Ax\|_v \leq f(A) \cdot \|x\|_v$.
- There exists a vector x, such that $\|Ax\|_v = f(A) \cdot \|x\|_v$.
- Then the value $f(A)$ is indeed the matrix norm induced by the vector norm $\|.\|_v$, so denote it by: $\|A\|_v$.

Proof for $\|.\|_1$: $\|A\|_1 = \max_{j=1}^{n} \sum_{i=1}^{n} |a_{ij}|$.

$$\|Ax\|_1 = \sum_{i=1}^{n} |(Ax)_i| = \sum_{i=1}^{n} \left| \sum_{j=1}^{n} a_{ij} x_j \right| \leq \sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}| \cdot |x_j| =$$

$$= \sum_{j=1}^{n} \left(|x_j| \cdot \underbrace{\sum_{i=1}^{n} |a_{ij}|} \right) \leq \underbrace{\left(\max_{j=1}^{n} \sum_{i=1}^{n} |a_{ij}| \right)} \cdot \|x\|_1 .$$

Let $x = e_k$, if the column sum is maximal for the kth column. Then

$$\|Ae_k\|_1 = \underbrace{\ldots} \underbrace{\|e_k\|_1} .$$

Natural matrix norms, induced norms

Proof outline for $\|.\|_\infty$: $$\|A\|_\infty = \max_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|.$$

- The estimation is very similar, like in case of $\|.\|_1$.
- Choose the vector
$$x = \begin{pmatrix} \pm 1 \\ \vdots \\ \pm 1 \end{pmatrix}$$
for equality with appropriately chosen signs...

Proof outline for $\|.\|_2$: $$\|A\|_2 = \left(\max_{i=1}^{n} \lambda_i(A^\top A)\right)^{1/2}.$$

- First prove that the eigenvalues are non-negative.
- Use diagonalization for the estimation.
- Choose the eigenvector for the largest eigenvalue for equality.

Proof for $\|.\|_2$ (skipped): read if interested.

First show that $A^\top A$ is symmetric and its eigenvalues are non-negative (i.e. $A^\top A$ is positive semidefinite).

- $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$, i.e. $A^\top A$ is symmetric, thus the eigenvalues are real (not complex).
- Let $y \neq 0$ an eigenvector of $A^\top A$ for the eigenvalue λ, i.e.

$$A^\top A y = \lambda \cdot y.$$

Multiply from the left by the vector $y^\top$:

$$y^\top A^\top A y = \lambda \cdot y^\top y.$$

$$\lambda = \frac{y^\top A^\top A y}{y^\top y} = \frac{(Ay)^\top (Ay)}{y^\top y} = \frac{\|Ay\|_2^2}{\|y\|_2^2} \geq 0.$$

Proof for $\|.\|_2$ (skipped): read if interested, continued.

Then following the definition, examine the norm of Ax.

We'll use that $A^\top A$ is symmetric, thus there exists an orthogonal matrix U (c.f. linear algebra), such that

$$A^\top A = U^\top DU \quad \Leftrightarrow \quad UA^\top AU^\top = D$$

with the eigenvalues of $A^\top A$ in the diagonal (and being non-negative). Introduce the notation $y = Ux$.

$$\|Ax\|_2^2 = (Ax)^\top (Ax) = x^\top A^\top Ax = x^\top U^\top DUx = (Ux)^\top D(Ux)$$
$$= y^\top Dy = \sum_{i=1}^{n} \underbrace{d_{ii}}_{\geq 0} \cdot |y_i|^2 \leq \max_{i=1}^{n} d_{ii} \cdot \sum_{i=1}^{n} |y_i|^2 = \max_{i=1}^{n} \lambda_i(A^\top A) \cdot \|y\|_2^2 \,.$$

We claim that $\|y\|_2^2 = \|x\|_2^2$.

Proof for $\|.\|_2$ (skipped): read if interested, continued.

Indeed $\|y\|_2^2 = y^\top y = (Ux)^\top (Ux) = x^\top U^\top U x = x^\top x = \|x\|_2^2$, thus

$$\|Ax\|_2^2 \leq \ldots \leq \max_{i=1}^{n} \lambda_i(A^\top A) \cdot \|x\|_2^2 \,.$$

For $x \neq 0$:

$$\frac{\|Ax\|_2}{\|x\|_2} \leq \left(\max_{i=1}^{n} \lambda_i(A^\top A) \right)^{1/2} .$$

It is left to prove that there exists such vector $x \neq 0$ where the supremum is taken, i.e. equality holds.

Let $\lambda_m = \max \lambda_i(A^\top A)$ and $v_m \neq 0$, $\|v_m\|_2 = 1$ is the corresponding eigenvector.

$$\|Av_m\|_2^2 = (Av_m)^\top (Av_m) = v_m^\top \underbrace{A^\top A\, v_m}_{\lambda_m \cdot v_m} = \lambda_m \cdot \underbrace{v_m^\top v_m}_{=1} = \lambda_m.$$

$\square$

Definítion: spectral radius

The *spectral radius* of the matrix $A \in \mathbb{R}^{n\times n}$ is $\varrho(A) := \max_{i=1}^{n} |\lambda_i(A)|$.

Remark. The spectral norm can be formulated using the spectral radius:

$$\|A\|_2 = \sqrt{\varrho(A^\top A)}.$$

Proposition

For a symmetric matrix $A \in \mathbb{R}^{n\times n}$

$$\|A\|_2 = \varrho(A).$$

Proof. Trivially from the properties of eigenvalues. $\square$

Example

Find the $\|.\|_1$ and $\|.\|_\infty$ norms of the below matrix.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 2 \end{bmatrix}$$

$$\|A\|_1 = \max\{1+2, |-4|+2\} = 6$$

$$\|A\|_\infty = \max\{1+|-4|, 2+2\} = 5$$

Example

Find the $\|.\|_2$ of the below matrix.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 2 \end{bmatrix}$$

$$A^\top A = \begin{bmatrix} 1 & 2 \\ -4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 20 \end{bmatrix},$$

In this case the eigenvalues are easily identified...

$$\|A\|_2 = \left(\max_{i=1}^{n} \lambda_i(A^\top A)\right)^{1/2} = \sqrt{\max\{5, 20\}} = \sqrt{20} \approx 4.4721.$$

Proposition

The Frobenius norm is not a natural matrix norm.

Proof. Consider the norms of the unit matrix $I \in \mathbb{R}^{n \times n}$.

- For induced norms $\|I\| = \sup_{x \neq 0} \frac{\|Ix\|_v}{\|x\|_v} = 1$.
- But $\|I\|_F = \sqrt{n}$.
- So there is no vector norm, which induces the Frobenius norm (if $n > 1$). $\square$

Further properties of matrix norms

Proposition: spectral radius and matrix norm

$$\varrho(A) \leq \|A\|$$

Proof. Prove that $|\lambda| \leq \|A\|$.
(For all λ and $v \neq 0$ eigenpairs.)

$$Av = \lambda v$$

$$Avv^\top = \lambda vv^\top$$

$$\|A\| \cdot \left\|vv^\top\right\| \geq \left\|Avv^\top\right\| = \left\|\lambda vv^\top\right\| = |\lambda| \cdot \left\|vv^\top\right\|$$

Divide by $\left\|vv^\top\right\| \neq 0$ to get $\|A\| \geq |\lambda|$. □

Further properties of matrix norms

Exercise: further properties

a. If $A^\top A = AA^\top$ (A is normal), then $\|A\|_2 = \varrho(A)$.
   Spec.: If A is symmetric, then it is normal.
b. If Q is orthogonal, then
   - $\|Qx\|_2 = \|x\|_2$,
   - $\|Q\|_2 = 1$,
   - $\|QA\|_2 = \|AQ\|_2 = \|A\|_2$.
c. $\|A\|_F^2 = \mathrm{tr}\,(A^\top A)$, with $\mathrm{tr}\,(B) := \sum_{k=1}^{n} b_{kk}$ (the 'trace' of B).
d. If Q is orthogonal, then $\|QA\|_F = \|AQ\|_F = \|A\|_F$.
e. $\|A\|_F^2 = \sum_{i=1}^{n} \lambda_i(A^\top A)$.
f. $\|.\|_F$ and $\|.\|_2$ are equivalent matrix norms.
g. The Frobenius norm is consisent with the Euclidean norm.

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Definition: condition number of a matrix

Given an invertible matrix $A \in \mathbb{R}^{n \times n}$ and matrix norm $\|.\|$, the value $\mathrm{cond}(A) := \|A\| \cdot \|A^{-1}\|$ is called the *condition number* of matrix A. (Denoted sometimes with $\kappa(A)$. [kappa])

Remark:

- Only defined for invertible matrices.
- The value depends on the choice of the norm.
  (E.g. $\mathrm{cond}_1(A), \mathrm{cond}_2(A), \dots$)

Proposition: properties of the condition number – part 1

- (a) $\text{cond}(A) \geq 1$ for natural matrix norms.
- (b) $\text{cond}(c \cdot A) = \text{cond}(A), \quad (c \in \mathbb{R}, c \neq 0).$
- (c) If Q is orthogonal, then $\text{cond}_2(Q) = 1$.

Proof.

- (a) $1 = \|I\| = \|A \cdot A^{-1}\| \leq \|A\| \cdot \|A^{-1}\| = \text{cond}(A).$
- (b) $\text{cond}(cA) = \|cA\| \cdot \|(cA)^{-1}\| = \|cA\| \cdot \left\|\frac{1}{c} A^{-1}\right\| =$

  $$= |c| \cdot \|A\| \cdot \frac{1}{|c|} \cdot \|A^{-1}\| = \text{cond}(A).$$

- (c) $\|Q\|_2 = \sup\limits_{x \neq 0} \dfrac{\|Qx\|_2}{\|x\|_2} = \sup\limits_{x \neq 0} \dfrac{\sqrt{x^\top Q^\top Q x}}{\sqrt{x^\top x}} = 1$

  $\|Q^{-1}\|_2 = \left\|Q^\top\right\|_2 = 1, \quad \text{cond}_2(Q) = 1$ $\square$

Proposition: properties of the condition number – part 2

- **d** If A is symmetric, then $\operatorname{cond}_2(A) = \dfrac{\max|\lambda_i(A)|}{\min|\lambda_i(A)|}$.
- **e** If A is symm., positive definite, then $\operatorname{cond}_2(A) = \frac{\max \lambda_i(A)}{\min \lambda_i(A)}$.
- **f** If A is invertible, then $\operatorname{cond}(A) \geq \frac{\max|\lambda_i(A)|}{\min|\lambda_i(A)|}$.

Proof.

- **d** Recall: $\|A\|_2 = \sqrt{\max \lambda_i(A^\top A)}$.
  But $\lambda_i(A^\top A) = \lambda_i(A^2) = (\lambda_i(A))^2$, so $\|A\|_2 = \max|\lambda_i(A)|$.
  For the inverse: $\|A^{-1}\|_2 = \max|\lambda_i(A^{-1})| = \dfrac{1}{\min|\lambda_i(A)|}$.
- **e** No absolute value needed because of the positive definiteness.
- **f** $\|A\| \geq \varrho(A) = \max|\lambda_i(A)|$,
  $\|A^{-1}\| \geq \varrho(A^{-1}) = \frac{1}{\min|\lambda_i(A)|}$. $\square$

Contents

Perturbation of the vector

$$A \cdot x = b$$

Examine, how the solution changes, when the right-hand-side, i.e. the vector is *slightly* changed, 'perturbed'.
(Inaccurate measurements, round-off error, ...)

1. **Original:**
   given A and b, we can calculate the solution: x.
   $Ax = b$
2. **Modified:**
   given A and $b + \Delta b$, we can calculate the solution: $x + \Delta x$.
   $A(x + \Delta x) = (b + \Delta b)$

Obviously the solution will be also *slightly* different... (?)

Example.

1. Original:

$$\begin{bmatrix} 4.1 & 2.8 \\ 9.7 & 6.6 \end{bmatrix} \cdot x = \begin{bmatrix} 4.1 \\ 9.7 \end{bmatrix} \quad \rightarrow \quad \text{solution: } x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

2. Modified:

$$\begin{bmatrix} 4.1 & 2.8 \\ 9.7 & 6.6 \end{bmatrix} \cdot (x + \Delta x) = \begin{bmatrix} 4.11 \\ 9.7 \end{bmatrix}$$

3.

$$\text{solution of the modified system:} \quad x + \Delta x = \begin{bmatrix} 0.34 \\ 0.97 \end{bmatrix}$$

Perturbation of the vector

How can we describe the change in the solution compared to the change in the right-hand-side?

- Change in the right-hand-side: $\delta b := \dfrac{\|\Delta b\|}{\|b\|} = 9.4959 \cdot 10^{-4}$.
- The change in the solution: $\delta x := \dfrac{\|\Delta x\|}{\|x\|} = 1.1732$.
- Their ratio: $\dfrac{\delta x}{\delta b} = 1235.5$.
- $\mathrm{cond}(A) = 1623$

Remark. Terminology: ill-conditioned, well-conditioned systems.

Theorem: sensitivity of the SLE
with respect to the perturbation of the vector

If A is invertible and $b \neq 0$, then using consistent norms

$$\frac{1}{\|A\| \cdot \|A^{-1}\|} \cdot \frac{\|\Delta b\|}{\|b\|} \leq \frac{\|\Delta x\|}{\|x\|} \leq \|A\| \cdot \left\|A^{-1}\right\| \cdot \frac{\|\Delta b\|}{\|b\|},$$

i.e.

$$\frac{1}{\mathrm{cond}(A)} \cdot \delta b \leq \delta x \leq \mathrm{cond}(A) \cdot \delta b.$$

Proof.

1. Subtract $Ax = b$ from $A(x + \Delta x) = (b + \Delta b)$, so $A\Delta x = \Delta b$.
2. Thus also $x = A^{-1}b$ and $\Delta x = A^{-1}\Delta b$ hold.

Proof. (continued)

3. So we have 4 equations:

$$b = Ax, \;\; x = A^{-1}b, \;\; \Delta b = A\Delta x, \;\; \Delta x = A^{-1}\Delta b.$$

4. Take norms for each. (Using consistent norms.)
   a. $\|b\| = \|Ax\| \quad \Rightarrow \quad \|b\| \le \|A\| \cdot \|x\| \quad \Rightarrow \quad \|x\| \ge \dfrac{\|b\|}{\|A\|}$,
   b. $\|\Delta b\| = \|A\Delta x\| \;\; \Rightarrow \;\; \|\Delta b\| \le \|A\| \cdot \|\Delta x\| \Rightarrow \|\Delta x\| \ge \dfrac{\|\Delta b\|}{\|A\|}$,
   c. $\|x\| = \left\|A^{-1}b\right\| \quad \Rightarrow \quad \|x\| \le \left\|A^{-1}\right\| \cdot \|b\|$,
   d. $\|\Delta x\| = \left\|A^{-1}\Delta b\right\| \quad \Rightarrow \quad \|\Delta x\| \le \left\|A^{-1}\right\| \cdot \|\Delta b\|$.

Proof. (continued)

5. Lower estimate using (b) and (c):

$$\frac{\|\Delta x\|}{\|x\|} \geq \frac{\frac{\|\Delta b\|}{\|A\|}}{\|A^{-1}\| \cdot \|b\|} = \frac{1}{\|A\| \cdot \|A^{-1}\|} \cdot \frac{\|\Delta b\|}{\|b\|}.$$

6. Upper estimate using (a) and (d) :

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\|A^{-1}\| \cdot \|\Delta b\|}{\frac{\|b\|}{\|A\|}} = \|A\| \cdot \left\|A^{-1}\right\| \cdot \frac{\|\Delta b\|}{\|b\|}.$$

$\square$

Perturbation of the matrix

$$A \cdot x = b$$

Examine, how the solution changes, when the left-hand-side, i.e. the matrix is *slightly* changed, 'perturbed'.
(Inaccurate measurements, round-off error, . . .)

1. **Original:**
   given A and b, we can calculate the solution: x.
   $Ax = b$
2. **Modified:**
   given $A + \Delta A$ and b, we can calculate the solution: $x + \Delta x$.
   $(A + \Delta A)(x + \Delta x) = b$

Obviously the solution will be also *slightly* different. . . (?)

Example.

1. Original:

$$\begin{bmatrix} 4.1 & 2.8 \\ 9.7 & 6.6 \end{bmatrix} \cdot x = \begin{bmatrix} 4.1 \\ 9.7 \end{bmatrix} \quad \to \quad \text{solution: } x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

2. Modified:

$$\begin{bmatrix} 4.11 & 2.8 \\ 9.7 & 6.6 \end{bmatrix} \cdot (x + \Delta x) = \begin{bmatrix} 4.1 \\ 9.7 \end{bmatrix}$$

3.

$$\text{solution of the modified system:} \quad x + \Delta x = \begin{bmatrix} 2.94 \\ -2.85 \end{bmatrix}$$

Perturbation of the matrix

How can we describe the change in the solution compared to the change in the left-hand-side?

- Change in the matrix: $\delta A := \dfrac{\|\Delta A\|}{\|A\|} = 7.8495e - 004.$
- The change in the solution: $\delta x := \dfrac{\|\Delta x\|}{\|x\|} = 3.4507.$
- Their ratio: $\dfrac{\delta x}{\delta A} = 4396.1.$
- $\mathrm{cond}(A) = 1623$

Theorem: sensitivity of the SLE
with respect to the perturbation of the matrix

If A is invertible, $b \neq 0$ and $\|\Delta A\| \cdot \|A^{-1}\| < 1$, then using induced norms

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\|A\| \cdot \|A^{-1}\|}{1 - \|\Delta A\| \cdot \|A^{-1}\|} \cdot \frac{\|\Delta A\|}{\|A\|}.$$

Lemma

If $\|M\| < 1$, then $(I + M)$ is invertible and using induced norms

$$\left\|(I + M)^{-1}\right\| \leq \frac{1}{1 - \|M\|}.$$

Remark. Induced norm is needed for the proof of the lemma.

Proof. Lemma.

- The matrix $I + M$ is invertible, since $\varrho(M) \leq \|M\| < 1$, i.e. for the eigenvalues of M: $|\lambda_i(M)| < 1$, i.e. are inside the unit circle. Verify that the eigenvectors for $I + M$ and M are the same, and for the eigenvalues $\lambda_i(I + M) = 1 + \lambda_i(M)$ holds. Thus 0 can not be an eigenvalue of $I + M$, thus $I + M$ is invertible.

- Examine the inverse of $I + M$, then its norm.

$$(I + M)^{-1} = I \cdot (I + M)^{-1} = (I + M - M)(I + M)^{-1} =$$
$$= I - M \cdot (I + M)^{-1},$$
$$\left\|(I + M)^{-1}\right\| \leq \|I\| + \|M\| \cdot \left\|(I + M)^{-1}\right\|,$$
$$(1 - \|M\|) \cdot \left\|(I + M)^{-1}\right\| \leq \|I\| = 1 \quad \Rightarrow \quad \left\|(I + M)^{-1}\right\| \leq \frac{1}{1 - \|M\|}.$$

$\square$

Proof. Theorem. Subtract $Ax = b$ from $(A + \Delta A)(x + \Delta x) = b$ to get $(A + \Delta A) \cdot \Delta x + \Delta A \cdot x = 0$, i.e.

$$(A + \Delta A) \cdot \Delta x = -\Delta A \cdot x,$$
$$A \cdot (I + A^{-1} \cdot \Delta A) \cdot \Delta x = -\Delta A \cdot x.$$

$\|A^{-1} \cdot \Delta A\| \le \|A^{-1}\| \cdot \|\Delta A\| < 1$ was assumed, so using the lemma we can say that $(I + A^{-1} \cdot \Delta A)$ is invertible.

$$\Delta x = -(I + A^{-1} \cdot \Delta A)^{-1} A^{-1} \Delta A \cdot x$$

Using our estimate for the norm of the inverse:

$$\|\Delta x\| \le \left\|(I + A^{-1} \cdot \Delta A)^{-1}\right\| \cdot \left\|A^{-1}\right\| \cdot \|\Delta A\| \cdot \|x\|$$

$$\frac{\|\Delta x\|}{\|x\|} \le \frac{1}{1 - \|A^{-1} \cdot \Delta A\|} \cdot \left\|A^{-1}\right\| \cdot \|\Delta A\| \le \frac{\|A\| \cdot \|A^{-1}\|}{1 - \|A^{-1}\| \cdot \|\Delta A\|} \cdot \frac{\|\Delta A\|}{\|A\|}.$$

□

Alternative forms:

$$\frac{\|A\| \cdot \|A^{-1}\|}{1 - \|\Delta A\| \cdot \|A^{-1}\|} \cdot \frac{\|\Delta A\|}{\|A\|} =$$

$$= \frac{\|\Delta A\| \cdot \|A^{-1}\|}{1 - \|\Delta A\| \cdot \|A^{-1}\|} =$$

$$= \frac{\|A\| \cdot \|A^{-1}\|}{1 - \frac{\|\Delta A\|}{\|A\|} \cdot \|A\| \cdot \|A^{-1}\|} \cdot \frac{\|\Delta A\|}{\|A\|} =$$

$$= \frac{\operatorname{cond}(A)}{1 - \operatorname{cond}(A) \cdot \frac{\|\Delta A\|}{\|A\|}} \cdot \frac{\|\Delta A\|}{\|A\|}.$$

Remark. Unified perturbation theorem

If we consider the system

$$A \cdot x = b$$

and both the matrix on the left-hand-side and the vector on the right-hand-side is subject to change, and for the solution of the perturbed system

$$(A + \Delta A) \cdot (x + \Delta x) = b + \Delta b$$

holds, then the following estimate can be proven:

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\mathrm{cond}(A)}{1 - \mathrm{cond}(A) \cdot \frac{\|\Delta A\|}{\|A\|}} \cdot \left(\frac{\|\Delta A\|}{\|A\|} + \frac{\|\Delta b\|}{\|b\|} \right).$$

The effect of LU decomposition on the sensitivity

Example

How does the LU decomposition influence the sensitivity of the system? Show that it does not get better.

Proof.

- $Ax = b \quad \Rightarrow \quad LUx = b \quad \Rightarrow \quad Ly = b, Ux = y,$
- $A = L \cdot U \quad \Rightarrow \quad \|A\| \leq \|L\| \cdot \|U\|$
- $A^{-1} = U^{-1} \cdot L^{-1} \quad \Rightarrow \quad \|A^{-1}\| \leq \|L^{-1}\| \cdot \|U^{-1}\|$
- $\mathrm{cond}(A) \leq \mathrm{cond}(L) \cdot \mathrm{cond}(U)$ □

Furthermore it can happen that $\mathrm{cond}(L), \mathrm{cond}(U) >> \mathrm{cond}(A)$, i.e. for some matrices the Gaussian elimination may give the result with quite bad precision. (Cholesky and QR are better.)

Contents

The condition number describes the sensitivity of the SLE (i.e. the problem), not the method for the solution (i.e. the algorithm). For that purpose, the residual vector is used.

Definition: residual vector

Let $\widetilde{x}$ denote an approximate solution of the system $Ax = b$. Then the vector $r := b - A\widetilde{x}$ is called the *residual* or *remainder vector*.

Remark. Easy to calculated. Also used in iterative methods.

Relative residual

Definition: relative residual

- The value $\eta := \frac{\|r\|}{\|A\| \cdot \|\widetilde{x}\|}$ ([eta]) is called the *relative residual*.
- By the definition of stability (inverse direction) the method is stable, if the system $(A + \Delta A) \cdot \widetilde{x} = b$ corresponding to the approximate solution $\widetilde{x}$ is just slightly perturbed compared to the original system, i.e. $\frac{\|\Delta A\|}{\|A\|}$ is small.

η is easy to calculate knowing the approximate solution.
We want to give estimates for $\frac{\|\Delta A\|}{\|A\|}$ without knowing ΔA.

Theorem: estimation for the relative residual

If A is invertible, then using consistent norms $\eta \leq \dfrac{\|\Delta A\|}{\|A\|}$,

i.e. if η is large, then $\frac{\|\Delta A\|}{\|A\|}$ is also large.

Proof. $b = (A + \Delta A) \cdot \widetilde{x} = A \cdot \widetilde{x} + \Delta A \cdot \widetilde{x}$, thus
$b - A \cdot \widetilde{x} = r = \Delta A \cdot \widetilde{x}$. Using the consistency of the norms

$$\|r\| \leq \|\Delta A\| \cdot \|\widetilde{x}\| .$$

$$\eta = \frac{\|r\|}{\|A\| \cdot \|\widetilde{x}\|} \leq \frac{\|\Delta A\| \cdot \|\widetilde{x}\|}{\|A\| \cdot \|\widetilde{x}\|} \leq \frac{\|\Delta A\|}{\|A\|}.$$

$\square$

Theorem: relative residual in 2-norm

If A is invertible, then $\eta_2 = \dfrac{\|\Delta A\|_2}{\|A\|_2}$.

Proof. We show that

$$\Delta A = \frac{r\widetilde{x}^\top}{\widetilde{x}^\top \widetilde{x}}$$

is good for a perturbation, i.e. $\widetilde{x}$ is an exact solution for a SLE changed by this matrix. Substitute:

$$(A + \Delta A) \cdot \widetilde{x} = \left(A + \frac{r\widetilde{x}^\top}{\widetilde{x}^\top \widetilde{x}} \right) \cdot \widetilde{x} =$$

$$= A\widetilde{x} + \frac{r\widetilde{x}^\top \widetilde{x}}{\widetilde{x}^\top \widetilde{x}} = A\widetilde{x} + (b - A\widetilde{x}) = b.$$

Proof. (continued) We use $\left\| r\widetilde{x}^\top \right\|_2 = \|r\|_2 \cdot \|\widetilde{x}\|_2$.
(May be submitted.)

Estimating the relative residual

$$\frac{\|\Delta A\|_2}{\|A\|_2} = \frac{\left\| r\widetilde{x}^\top \right\|_2}{\|A\|_2 \|\widetilde{x}\|_2^2} = \frac{\|r\|_2 \|\widetilde{x}\|_2}{\|A\|_2 \|\widetilde{x}\|_2^2} = \frac{\|r\|_2}{\|A\|_2 \|\widetilde{x}\|_2} = \eta_2.$$

$\square$

If η_2 is small, then $\frac{\|\Delta A\|_2}{\|A\|_2}$ is small.
If $\eta_2 < \varepsilon_1$, then it is not possible to give a more precise solution in the given (machine) arithmetic.

Examples in Matlab

1. Induced norms for $\mathbb{R}^2$, $p = 2$.
2. Approximating induced norms for arbitrary $\mathbb{R}^n$ and p (trying $m = 100, \dots, 1000$ vectors).
3. Ill-conditioned and well-conditioned matrices.
4. Perturbations of linear systems.